

Phase 0 — Session Status Report

Date: 2026-06-22

Session: 006 (Phase 0 empirical pipeline)

Model / interface: Claude Opus 4.8 via Claude Code CLI

Scope: Phase 0 only (asset universe), per 04_code/CLAUDE_CODE_KICKOFF_PROMPT.md and 04_code/DATA_SPECIFICATION.md Section 7.

Status: ■ Phase 0 complete. Stopped for review before Phase 1, as instructed.

1. What was built

All scripts are reproducible and re-runnable; raw API payloads are cached so re-runs don't re-hit the network.

File	Purpose
04_code/fetch_cmc_snapshots.py	Pulls CMC historical top-1000 listings for each month-end 2015-08→2026-05; caches raw JSON to 03_data/raw/cmc_snapshots/ (130 files).
04_code/build_universe.py	Applies the Section 2.1 inclusion rule; emits the membership panel + assets table + stablecoin-exclusion list + rank sensitivity.
04_code/classify_assets.py	Functional coin/token/other classification with evidence, ambiguity flags, and ETH's dated PoW→PoS transition.
04_code/coverage_report.py	Generates the coverage/feasibility report + by-month/by-year CSVs.

Data outputs (03_data/): universe_panel.csv / .parquet (156,838 asset-months), universe_assets.csv (1,939 qualifying assets), universe_stablecoins_excluded.csv (71), universe_rank_sensitivity.csv, classification_table.csv (2,010 incl. excluded stablecoins), coverage_by_month.csv, coverage_by_year.csv, PHASE0_COVERAGE_REPORT.md.

Logs: 04_code/DATA_DECISIONS_LOG.md (9 entries appended);

06_documentation/ai_conversations/session_006_2026-06-22_phase0_pipeline.md;

06_documentation/time_log.md updated.

2. Key feasibility findings (verified live, not assumed)

- **CoinGecko public API caps historical data at the past 365 days** → unusable as the historical-ranking backbone on the free tier.
- **CoinMarketCap free historical listings data-api** serves daily, point-in-time ranked snapshots back to ~2013, **including delisted coins** → adopted as the Phase 0 backbone (this is what makes the universe survivorship-bias-free).
- **DeFiLlama** free API works → used for governance-token classification evidence.

- **Artemis** has no accessible free API (410 / DNS failure) → deferred.
-

3. What the coverage report shows

- **Live (observed) cross-section:** ~290 assets/month in 2015 rising to ~540–630 in 2020+. The observed top-250 count runs ~217–247/month (short of 250 because excluded stablecoins occupy real rank slots).
 - **Class mix shifts exactly as the theory section anticipated:** observed governance **tokens** grow from ~19 (2015) → ~139 (2020) → ~233 (2025); **coins** dominate the early sample (196 of 288 observed in 2015).
 - **Testability gates:** H1a (coins) has usable density from early in the sample; H1b / H2 / H3 (governance/voting-dependent) only become meaningfully testable from ~2020–2021. Treat earlier H1b/H2/H3 as out-of-support.
 - **Classification:** of 1,939 distinct names, 618 coin / 447 token / 874 other. The "other" bucket is large by raw name count (dead microcap tail, oracle/exchange/wrapped tokens) but small weighted by *observed asset-months* — the persistent, higher-cap assets that drive the tests are predominantly coin/token, and all spot-checked majors classify correctly.
-

4. Judgment calls made this session (all logged)

1. CoinGecko → CoinMarketCap as the ranking source (human-approved).
 2. Stablecoins excluded entirely from the universe, ranking-inclusive (human decision).
 3. N=250 default; sensitivity at 200/250/300 reported.
 4. Carry-forward (with last-observed price + status flag) for in-panel-but-unobserved months; death-return treatment deferred to Phase 3.
 5. Tag-rules + DeFiLlama + a curated 22-id native-coin override for classification; every non-clean case flagged `ambiguous_flag=True`.
 6. ETH treated as a dated PoW→PoS transition (staking from 2020-12-01, Merge 2022-09-15).
-

5. Open questions / decisions needed before Phase 1

1. **Carry-forward death returns:** how should assets that fall below rank 1000 be treated in return construction (final delisting return vs. constant carry)? (Phase 3, but decide the policy before λ /return work.)
2. **'other'/ambiguous classifications:** a manual confirmation pass is needed for any other/ambiguous asset that will actually enter the tests.
3. **Confirm N=250** vs. a tighter screen once λ -channel coverage is known.

4. **Artemis / paid CoinGecko:** obtain access if their fundamentals are wanted for the PQ series in Phase 2 (no free Artemis API; CoinGecko Pro key would enable the two-source market-cap cross-check the spec asks for).

No Phase 1 (λ channels) or later work was started, per the kickoff instruction.